

DAF26TZ06-NV007 - Automated Combat Assessment at the Edge

USAF (AFRL) | STTR | Release 6 | Opens Sep 23, 2026 (TPOC direct contact ends) | DSIP Q&A closes Oct 7, 12:00 ET | Proposals due Oct 21, 12:00 ET

Recommendation: PURSUE with a remote-sensing research partner. An edge-ML damage-classification problem with a public benchmark dataset that maps almost exactly onto the ask. STTR requires a research institution; ITAR applies. Confidence MEDIUM-HIGH.

Phase I award	\$300,000, 6 months
Proposal format	Technical volume 20 pages
Compliance	ITAR/EAR clause (foreign nationals restricted). CMMC Level 2 (Self). STTR: RI >= 30%, small business >= 40%.
Phase I deliverable	Analysis and design of the system with preliminary laboratory model testing; feasibility findings and a Phase II plan
Phase II	Functional prototype software + model weights on representative edge hardware, ICDs, demo in a simulated or operationally relevant environment

What they want

[TOPIC TEXT] "Awardees must demonstrate an innovative approach to physical and functional combat assessment, and novel or under-utilized data modalities."

Three components: edge-processing software that runs on sensors/UAVs under low SWaP and contested comms; AI/ML damage classification (physical and functional) aligned to CJCSI 3162.02 with confidence intervals per ICD 203; and phased test and evaluation from lab to simulated to field.

The sponsor names the data problem itself: "the lack of real time pre- and post-event satellite imagery for model training." A proposal that solves the training-data problem credibly has an edge.

Phase III named uses: urban surveillance, disaster and humanitarian response.

How we would build it

- **Model:** lightweight change-detection/segmentation network on pre/post image pairs (building- and vehicle-level damage classes), distilled and quantized for Jetson/Coral-class boards.
- **Functional damage layer:** rule/graph reasoning on top of physical damage (e.g., a radar site is 'functionally destroyed' if antenna and power are hit) - this is the under-served half of the ask.
- **Novel modalities:** add SAR change detection (day/night, cloud) and thermal signatures; fuse with EO. Public SAR and thermal datasets exist to show feasibility.
- **Confidence reporting:** calibrated probabilities mapped to ICD 203 language (e.g., 'likely', 'highly likely'), with an evidence chain per assessment.
- **Edge/comms:** compact assessment messages, on-device queueing for intermittent links, and a bandwidth budget.
- **Phase I lab test:** accuracy/latency/power on the edge board against a held-out damage benchmark.

Resources we would use

- xBD / xView2 building-damage dataset (pre/post disaster imagery with damage grades) - the standard public benchmark.

- Public SAR archives (Sentinel-1, Capella open data) for change-detection experiments; FLIR-class public thermal sets.
- CJCSI 3162.02A (combat assessment methodology) and ICD 203 (analytic standards) - both public, both cited in the topic.
- COTS edge boards: NVIDIA Jetson Orin Nano, Coral Edge TPU, Hailo-8L.
- RI partner: a university remote-sensing / computer-vision lab (needed for STTR and for the 'novel modalities' credibility).
- AFRL BAA FA8750-25-S-7002 'Targeting Operations and Analytics Development' (cited) - read for the sponsor's framing.

Bottlenecks and risks

- **Domain gap:** disaster damage is not combat damage; military target types and functional-kill logic need SME input, and real military pre/post imagery is unavailable in Phase I.
- **STTR partner timing:** an RI agreement must be in place by Oct 21; start the conversation immediately.
- **ITAR:** staffing restriction on foreign nationals.
- **Edge constraints:** the topic wants low SWaP under contested comms - the model must actually run on the board, not just in the cloud; demonstrate this in Phase I.
- **Competition:** AFRL Rome's targeting/analytics ecosystem has incumbents; the differentiator must be the functional-assessment reasoning and the modality fusion, not a generic detector.

Decision factors

- Fit to team: edge ML pipelines - strong (same skill set as the Release 5 semantic-ISR topic).
- Dual use: disaster/humanitarian damage mapping is an existing commercial and NGO market.
- Decisive pre-proposal experiment: fine-tune a small segmentation model on xBD, quantize for a Jetson, and report accuracy, latency and watts - three numbers that would anchor the technical volume.

Before Oct 7 (Q&A): ask whether representative pre/post military-target imagery will be furnished in Phase II, which sensor modalities the sponsor considers 'under-utilized', and the target edge hardware class.

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